

Individual Reflection: Data Science Toolbox, Formative Sprint 0

Morgan L

Fit: area and goals

Our group chose sports analytics, with each member investigating a different resource or dataset. I focused on Premier League football by recreating an existing R analysis from the Akshat29-shah/epl-analysis GitHub repository, it compared the 2021/22, 2022/23 and 2023/24 seasons using match and event data, with particular focus to events such as goals, added time, cards and substitutions.

My first aim was to reproduce the author's results and assess how reproducible and reliable the analysis was, then I followed the investigation into whether the unusually high number of goals in 2023/24 was associated with increased added time, extended into future seasons to see if the trend continued and finally connected this to betting markets by using Football-Data match and bookmaker odds data through 2025/26.

Depth: results and mathematics

The reproduction was largely successful. Total goals increased from 1,071 in 2021/22 and 1,084 in 2022/23 to 1,246 in 2023/24, while mean goals per match rose to 3.28. Average second half added time also increased substantially, and the number of goals after the 90th minute rose from 49 in 2022/23 to 103 in 2023/24. However, only around one third of the increase in total goals between these seasons occurred after 90 minutes, so increased added time cannot explain the full rise in scoring.

One statistical issue I identified was the original use of a two-sample z-test for post-90 goals. The population standard deviations were not known; the values supplied to the test were estimated from the same samples. I therefore replaced this with Welch's two-sample t-test, unlike the equal variance t-test, it does not assume the two populations have the same variance. The test still gave strong evidence that the mean number of goals scored after 90 minutes was higher in 2023/24 than across the previous two seasons, with a one-sided p-value of approximately 0.00012.

I also extended the analysis using bookmaker Over 2.5 odds. For decimal odds o , the raw implied probability is $1/o$. Because bookmakers include a margin, I removed the two-way overround using

$$P(\text{Over } 2.5) = \frac{\left(\frac{1}{o_{\text{over}}}\right)}{\left(\frac{1}{o_{\text{over}}}\right) + \left(\frac{1}{o_{\text{under}}}\right)}$$

This allowed me to compare realised Over 2.5 frequencies with bookmaker expectations. The largest discrepancy occurred in 2023/24: 64.7% of matches finished Over 2.5 compared with an average market implied probability of about 59.1%. In later seasons the realised rate fell back, suggesting 2023/24 was unusually high scoring rather than clear evidence of a permanent trend.

Weaknesses. The original event data had unclear provenance and appeared to involve manual preprocessing, which made reproduction and extension difficult. Substitution comparisons

were invalidated by the change from three to five permitted substitutes from the 2021/22 season to the 2022/23 season. The analysis is observational, so associations between added time and goals should not be interpreted causally. Next time I would begin with a single documented raw data source and define the research question before reproducing the analysis.

Literature search and comfort zone

A substantial part of my time was spent searching for a suitable analysis to reproduce and extend. I looked through a range of Premier League and football data science projects, but many were unsuitable because the datasets were not publicly available, the research question was unclear, the code was incomplete, or the methods were too advanced to reproduce confidently within the time available. I found one promising analysis based on decision trees, but as we had not yet covered decision trees in the course, I decided not to use a method that I could run without properly understanding. This restricted my realistic choices mainly to techniques from my previous statistics background, including hypothesis testing, linear models and generalised linear models.

The GitHub analysis I eventually selected was therefore a compromise between methodological accessibility, available data and content. It had available R code and data, and a focus on changes in Premier League goals and match events across seasons, which gave me a suitable starting point for extension. However, the search also showed me that publicly available sports data projects vary considerably in quality and reproducibility. In particular, datasets were often poorly documented or had unclear preprocessing steps, which became an important limitation of the project I eventually chose.

My main comfort zone was statistical analysis and interpretation, particularly hypothesis testing, working with regression and statistical packages in R. I found data provenance and reproducibility more difficult, particularly understanding how the author's event files had been produced from the original sources. I would like more experience with reproducible data pipelines and with modelling football outcomes rather than only carrying out descriptive and inferential comparisons.

Collaboration and group working

GitHub allowed us to work in parallel on separate analyses and later combine them into one sports analytics project. However, because we used different datasets, languages and packages, our analyses were initially quite separate although my betting market extension helped connect my work more closely to Thomas's bookmaker analysis. Next time I would agree a common report structure, naming conventions and shared questions earlier so that the separate analyses connect more naturally.

Evidence

My contribution is the Premier League R Markdown analysis, including reproduction of the original GitHub project, corrections to the post-90 analysis and hypothesis test, additional visualisations, and the extension using 2024/25 and 2025/26 results and bookmaker odds.

References

[Akshat29-shah. epl-analysis. GitHub.](#)

[England Football Results Betting Odds | Premiership Results & Betting Odds](#)

<https://www.premierleague.com/en/news/4271974>

<https://www.premierleague.com/en/news/4580269/why-more-late-goals-are-being-scored-in-2025-26-premier-league-season>